

elementary antiderivative but $g(x)$ does not possess an elementary antiderivative. See Problems 31–34 in Exercises 2.2.

REMARKS

In some of the preceding examples we saw that the constant in the one-parameter family of solutions for a first-order differential equation can be relabeled when convenient. Also, it can easily happen that two individuals solving the same equation correctly arrive at dissimilar expressions for their answers. For example, by separation of variables, we can show that one-parameter families of solutions for the DE $(1 + y^2) dx + (1 + x^2) dy = 0$ are

$$\arctan x + \arctan y = c \quad \text{or} \quad \frac{x + y}{1 - xy} = c.$$

As you work your way through the next several sections, keep in mind that families of solutions may be equivalent in the sense that one family may be obtained from another by either relabeling the constant or applying algebra and trigonometry. See Problems 27 and 28 in Exercises 2.2.

2.2 Exercises

Answers to selected odd-numbered problems begin on page ANS-2.

In Problems 1–22, solve the given differential equation by separation of variables.

1. $\frac{dy}{dx} = \sin 5x$
2. $\frac{dy}{dx} = (x + 1)^2$
3. $dx + e^{3x} dy = 0$
4. $dy - (y - 1)^2 dx = 0$
5. $x \frac{dy}{dx} = 4y$

6. $\frac{dy}{dx} + 2xy^2 = 0$
7. $\frac{dy}{dx} = e^{3x+2y}$
8. $e^xy \frac{dy}{dx} = e^{-y} + e^{-2x-y}$
9. $y \ln x \frac{dx}{dy} = \left(\frac{y+1}{x}\right)^2$
10. $\frac{dy}{dx} = \left(\frac{2y+3}{4x+5}\right)^2$
11. $\csc y dx + \sec^2 x dy = 0$
12. $\sin 3x dx + 2y \cos^3 3x dy = 0$
13. $(e^y + 1)^2 e^{-y} dx + (e^x + 1)^3 e^{-x} dy = 0$
14. $x(1 + y^2)^{1/2} dx = y(1 + x^2)^{1/2} dy$
15. $\frac{dS}{dr} = kS$
16. $\frac{dQ}{dt} = k(Q - 70)$
17. $\frac{dP}{dt} = P - P^2$
18. $\frac{dN}{dt} + N = Nte^{t+2}$
19. $\frac{dy}{dx} = \frac{xy + 3x - y - 3}{xy - 2x + 4y - 8}$
20. $\frac{dy}{dx} = \frac{xy + 2y - x - 2}{xy - 3y + x - 3}$
21. $\frac{dy}{dx} = x\sqrt{1 - y^2}$
22. $(e^x + e^{-x}) \frac{dy}{dx} = y^2$

In Problems 23–30, find an explicit solution of the given initial-value problem.

23. $\frac{dx}{dt} = 4(x^2 + 1), x(\pi/4) = 1$

24. $\frac{dy}{dx} = \frac{y^2 - 1}{x^2 - 1}, y(2) = 2$
25. $x^2 \frac{dy}{dx} = y - xy, y(-1) = -1$
26. $\frac{dy}{dt} + 2y = 1, y(0) = \frac{5}{2}$
27. $\sqrt{1 - y^2} dx - \sqrt{1 - x^2} dy = 0, y(0) = \sqrt{3}/2$
28. $(1 + x^4) dy + x(1 + 4y^2) dx = 0, y(1) = 0$
29. $\frac{dy}{dx} = -y \ln y, y(0) = e$
30. $x \sinh y \frac{dy}{dx} = \cosh y, y(1) = 0$

In Problems 31–34, proceed as in Example 6 and find an explicit solution of the given initial-value problem.

31. $\frac{dy}{dx} = ye^{-x^2}, y(4) = 1$
32. $\frac{dy}{dx} = y^2 \sin x^2, y(-2) = \frac{1}{3}$
33. $\frac{dy}{dx} = (1 + y^2)\sqrt{1 + \cos x^3}, y(1) = 1$
34. $\frac{dy}{dx} = \frac{e^{-2y} \sin x}{1 + x^2}, y(0) = 0$

In Problems 35–38, find an explicit solution of the given initial-value problem. Determine the exact interval I of definition of each solution by analytical methods. Use a graphing utility to plot the graph of each solution.

35. $\frac{dy}{dx} = \frac{2x + 1}{2y}, y(-2) = -1$
36. $(2y - 2) \frac{dy}{dx} = 3x^2 + 4x + 2, y(1) = -2$
37. $e^y dx - e^{-x} dy = 0, y(0) = 0$
38. $\sin x dx + y dy = 0, y(0) = 1$
- 39.

- (a) Find a solution of the initial-value problem consisting of the differential equation in Example 4 and the initial conditions $y(0) = 2$, $y(0) = -2$, $y(\frac{1}{4}) = 1$.
- (b) Find the solution of the differential equation in Example 4 when $\ln c_1$ is used as the constant of integration on the *left-hand* side in the solution and $4 \ln c_1$ is replaced by $\ln c$. Then solve the same initial-value problems in part (a).
40. Find a solution of $x \frac{dy}{dx} = y^2 - y$ that passes through the indicated points.
- (a) $(0, 1)$
- (b) $(0, 0)$
- (c) $(\frac{1}{2}, \frac{1}{2})$
- (d) $(2, \frac{1}{4})$
41. Find a singular solution of Problem 21. Of Problem 22.
42. Show that an implicit solution of

$$2x \sin^2 y \, dx - (x^2 + 10) \cos y \, dy = 0$$

is given by $\ln(x^2 + 10) \csc y = c$. Find the constant solutions, if any, that were lost in the solution of the differential equation.

Often a radical change in the form of the solution of a differential equation corresponds to a very small change in either the initial condition or the equation itself. In Problems 43–46, find an explicit solution of the given initial-value problem. Use a graphing utility to plot the graph of each solution. Compare each solution curve in a neighborhood of $(0, 1)$.

43. $\frac{dy}{dx} = (y - 1)^2, \quad y(0) = 1$
44. $\frac{dy}{dx} = (y - 1)^2, \quad y(0) = 1.01$
45. $\frac{dy}{dx} = (y - 1)^2 + 0.01, \quad y(0) = 1$

46. $\frac{dy}{dx} = (y - 1)^2 - 0.01, \quad y(0) = 1$
47. Every autonomous first-order equation $dy/dx = f(y)$ is separable. Find explicit solutions $y_1(x)$, $y_2(x)$, $y_3(x)$, and $y_4(x)$ of the differential equation $dy/dx = y - y^3$ that satisfy, in turn, the initial conditions $y_1(0) = 2$, $y_2(0) = \frac{1}{2}$, $y_3(0) = -\frac{1}{2}$, and $y_4(0) = -2$. Use a graphing utility to plot the graphs of each solution. Compare these graphs with those predicted in Problem 19 of Exercises 2.1. Give the exact interval of definition for each solution.
- 48.
- (a) The autonomous first-order differential equation $dy/dx = 1/(y - 3)$ has no critical points. Nevertheless, place 3 on a phase line and obtain a phase portrait of the equation. Compute d^2y/dx^2 to determine where solution curves are concave up and where they are concave down (see Problems 35 and 36 in Exercises 2.1). Use the phase portrait and concavity to sketch, by hand, some typical solution curves.
- (b) Find explicit solutions $y_1(x)$, $y_2(x)$, $y_3(x)$, and $y_4(x)$ of the differential equation in part (a) that satisfy, in turn, the initial conditions $y_1(0) = 4$, $y_2(0) = 2$, $y_3(1) = 2$, and $y_4(-1) = 4$. Graph each solution and compare with your sketches in part (a). Give the exact interval of definition for each solution.

In Problems 49–54, use a technique of integration or a substitution to find an explicit solution of the given differential equation or initial-value problem.

49. $\frac{dy}{dx} = \frac{1}{1 + \sin x}$

50. $\frac{dy}{dx} = \frac{\sin \sqrt{x}}{\sqrt{y}}$
51. $(\sqrt{x} + x) \frac{dy}{dx} = \sqrt{y} + y$
52. $\frac{dy}{dx} = y^{2/3} - y$
53. $\frac{dy}{dx} = \frac{e^{\sqrt{x}}}{y}, \quad y(1) = 4$
54. $\frac{dy}{dx} = \frac{x \tan^{-1} x}{y}, \quad y(0) = 3$

Discussion Problems

- 55.
- (a) Explain why the interval of definition of the explicit solution $y = \phi_2(x)$ of the initial-value problem in Example 2 is the *open* interval $(-5, 5)$.
- (b) Can any solution of the differential equation cross the x -axis? Do you think that $x^2 + y^2 = 1$ is an implicit solution of the initial-value problem $dy/dx = -x/y, y(1) = 0$?
56. On page 48 we showed that a one-parameter family of solutions of the first-order differential equation $dy/dx = xy^{1/2}$ is $y = (\frac{1}{4}x^4 + c)^2$ for $c \geq 0$. Each solution in this family is defined on the interval $(-\infty, \infty)$. The last statement is not true if we choose c to be negative. For $c = -1$, explain why $y = (\frac{1}{4}x^4 - 1)^2$ is *not* a solution of the DE on $(-\infty, \infty)$. Find an interval of definition I on which $y = (\frac{1}{4}x^4 - 1)^2$ is a solution of the DE.
57. In Problems 47 and 48 we saw that every autonomous first-order differential equation $dy/dx = f(y)$ is separable. Does this fact help in the solution of the initial-value problem $\frac{dy}{dx} = \sqrt{1 + y^2} \sin^2 y, y(0) = \frac{1}{2}$? Discuss. Sketch, by hand, a

plausible solution curve of the problem.

58. (a) Solve the two initial-value problems:

$$\frac{dy}{dx} = y, y(0) = 1 \quad \text{and} \quad \frac{dy}{dx} = y + \frac{y}{x \ln x}, y(e) = 1.$$

(b) Show that there are more than 1.65 million digits in the y -coordinate of the point of intersection of the two solution curves in part (a).

59. Find a function whose square plus the square of its derivative is 1.
60. (a) The differential equation in Problem 27 is equivalent to the normal form

$$\frac{dy}{dx} = \sqrt{\frac{1 - y^2}{1 - x^2}}$$

in the square region in the xy -plane defined by $|x| < 1$, $|y| < 1$. But the quantity under the radical is nonnegative also in the regions defined by $|x| > 1$, $|y| > 1$. Sketch all regions in the xy -plane for which this differential equation possesses real solutions.

(b) Solve the DE in part (a) in the regions defined by $|x| > 1$, $|y| > 1$. Then find an implicit and an explicit solution of the differential equation subject to $y(2) = 2$.

Mathematical Models

61. **Hawking Radiation** According to the English theoretical physicist **Stephen Hawking** (1942–2018), the mass $m(t)$ of a black hole decreases over time due to radiation. A model for the mass is given by

$$\frac{dm}{dt} = -\frac{k}{m^2}.$$

Solve for $m(t)$ when $m(0) = m_0$.

- 62. Hanging Chain** Suppose a flexible steel chain is hanging under its own weight and its ends are attached at the same height. See the photo below.

(a) When determining the shape of the chain, one has to first solve the differential equation

$$\frac{du}{dx} = a\sqrt{1 + u^2},$$

where a is a constant. Solve for $u(x)$ when $u(0) = 0$.

(b) The function $u(x)$ found in part (a) is related to the shape of the chain $y(x)$ by the differential equation

$$\frac{dy}{dx} = u.$$

Solve for $y(x)$ when $y(0) = 1/a$. Identify the shape of the hanging chain.



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Hanging chain in Problem 62

- 63. Tautochrone Problem** The tautochrone problem consists of finding a curve C such that the time of descent of a bead sliding without friction on C to its lowest point is independent of its starting point. If the curve C in **FIGURE 2.2.6** is a tautochrone, then the three beads starting from different

positions corresponding to x -coordinates x_1 , x_2 , and x_3 will reach the origin $(0, 0)$ in the same time. The Dutch mathematician and astronomer **Christiaan Huygens** (1629–1695) was the first to publish the solution of the tautochrone problem in 1673.

In determining such a curve, the following differential equation must be solved:

$$1 + \left(\frac{dx}{dy} \right)^2 = \frac{a}{y},$$

where a is a constant. Find parametric equations for C by solving the DE in the form

$$dx = \sqrt{\frac{a - y}{y}} dy.$$

Use the substitution $y = a \sin^2 \phi$.

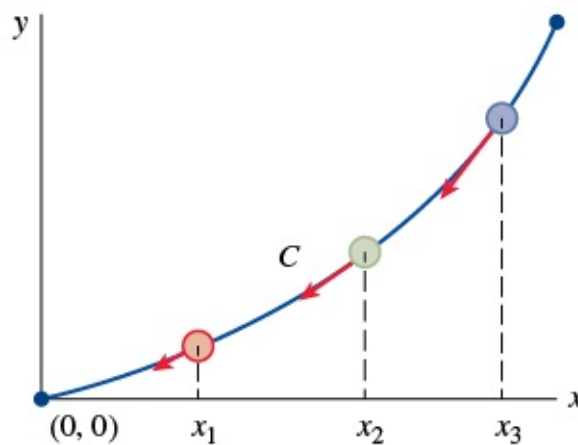


FIGURE 2.2.6 Tautochrone in Problem 63

- 64. Chemical Reaction** In a certain second-order chemical reaction involving a single reactant, the rate of reaction is given by

$$\frac{dx}{dt} = k(A - x)^2,$$

where k and A are constants and $x(t)$ is the number of grams of the new chemical formed in time t . Solve for $x(t)$ when $x(0) = 0$.

- 65. Suspension Bridge** In (16) of Section 1.3 we saw that a mathematical model for the shape of a flexible cable strung between two vertical supports is

$$\frac{dy}{dx} = \frac{W}{T_1}, \quad (11)$$

where W denotes the portion of the total vertical load between the points P_1 and P_2 shown in Figure 1.3.9. The DE (11) is separable under the following conditions that describe a suspension bridge.

Let us assume that the x - and y -axes are as shown in **FIGURE 2.2.7**—that is, the x -axis runs along the horizontal roadbed, and the y -axis passes through $(0, a)$, which is the lowest point on one cable over the span of the bridge, coinciding with the interval $[-L/2, L/2]$. In the case of a suspension bridge, the usual assumption is that the vertical load in (11) is only a uniform roadbed distributed along the horizontal axis. In other words, it is assumed that the weight of all cables is negligible in comparison to the weight of the roadbed and that the weight per unit length of the roadbed (say, pounds per horizontal foot) is a constant r . Use this information to set up and solve an appropriate initial-value problem from which the shape (a curve with equation $y = \phi(x)$) of each of the two cables in a suspension bridge is determined. Express your solution of the IVP in terms of the sag h and span L shown in Figure 2.2.7.

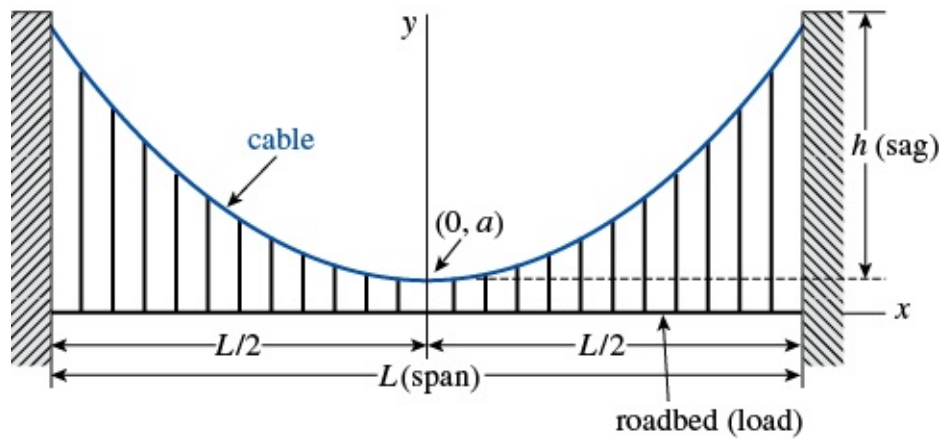


FIGURE 2.2.7 Shape of a cable in Problem 65

Computer Lab Assignments

66.

- (a) Use a CAS and the concept of level curves to plot representative graphs of members of the family of solutions of the differential equation $\frac{dy}{dx} = -\frac{8x + 5}{3y^2 + 1}$. Experiment with different numbers of level curves as well as various rectangular regions defined by $a \leq x \leq b$, $c \leq y \leq d$.
- (b) On separate coordinate axes plot the graphs of the particular solutions corresponding to the initial conditions: $y(0) = -1$; $y(0) = 2$; $y(-1) = 4$; $y(-1) = -3$.

67.

- (a) Find an implicit solution of the IVP

$$(2y + 2)dy - (4x^3 + 6x)dx = 0, y(0) = -3.$$

- (b) Use part (a) to find an explicit solution $y = \phi(x)$ of the IVP.
- (c) Consider your answer to part (b) as a *function* only. Use

a graphing utility or a CAS to graph this function, and then use the graph to estimate its domain.

- (d) With the aid of a root-finding application of a CAS, determine the approximate largest interval I of definition of the solution $y = \phi(x)$ in part (b). Use a graphing utility or a CAS to graph the solution curve for the IVP on this interval.

68.

- (a) Use a CAS and the concept of level curves to plot representative graphs of members of the family of solutions of the differential equation $\frac{dy}{dx} = \frac{x(1-x)}{y(-2+y)}$. Experiment with different numbers of level curves as well as various rectangular regions in the xy -plane until your result resembles **FIGURE 2.2.8**.
- (b) On separate coordinate axes, plot the graph of the implicit solution corresponding to the initial condition $y(0) = \frac{3}{2}$. Use a colored pencil to mark off that segment of the graph that corresponds to the solution curve of a solution ϕ that satisfies the initial condition. With the aid of a root-finding application of a CAS, determine the approximate largest interval I of definition of the solution ϕ . [Hint: First find the points on the curve in part (a) where the tangent is vertical.]
- (c) Repeat part (b) for the initial condition $y(0) = -2$.

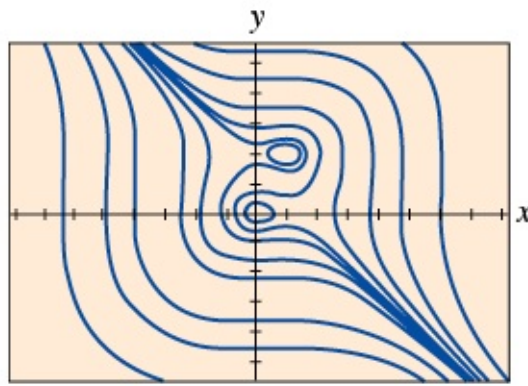


FIGURE 2.2.8 Level curves in Problem 68

*In Section 2.6 we discuss several other ways of proceeding that are based on the concept of a numerical solver.